COMP 3500SEF · Final Report

Warehouse Logistics Management System
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Testing: Chung Ngon Man, Li Mu Qiu.

1. Scope

Goals: traceable decisions and role-separated workflows. Roles: customer, staff, driver, administrator. Local-only: one app; seed data (RK-10, Table B).

2. Development process

Table A. Increments

Increment and date	Trigger / activity	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, responsibilities and API contracts recorded (register)
17 Oct review	Stock and order review	Two defects; fixed as row 3
Confirmation, 26–28 Oct 2026	Row-2 defects: repeat submits, no ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Role functions integrated	Customer, staff and admin workflows run end to end
Driver and privacy, 2 Nov 2026	Driver tests outstanding	Both pass (driver, privacy)
Diagram review, 10–11 Nov 2026	Diagrams diverged	Aligned (review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date export in three interactions (fill both dates, export)
CI health tests, 11–12 Nov 2026	No test covered CI-run outcomes	Two regression tests; suite reached 38 (unit.log)

3. Requirements

- FR-01 create orders with lines Test. - FR-02 enforce lifecycle, stock release Test (cancel). - FR-03 number orders, shipments, owners Test. - FR-04 inventory, ledger, low-stock alerts Test. - FR-05 record, read notifications Test. - FR-06 admin aggregates, CSV Test. - FR-07 authenticate, access Test. - FR-08 assign, deliver Test.

- NFR-01 performance: API p95 ≤2 s, first contentful paint p95 ≤3 s on the local host (measurement). - NFR-02 CI health — every commit ends in green unit, integration, browser runs (runs, observations). - NFR-03 least privilege: role gates; delivery address only to the assigned driver (privacy tests). - NFR-04 interaction budget: every core operation (the eleven in §8) takes at most three in-app interactions; one is a fill, selection or click (walkthrough). - NFR-05 process quality — every increment closes on a recorded review or test run.

Table B. Risk coverage

Risk	Addressed by	Evidence
①RK-01, RK-03, RK-06, RK-08 schedule, test week, coursework, video	weekly re-planning, scope priorities	risk plan
②RK-02 contract rework	lifecycle locked as test constants	constants tests
③RK-04, RK-07 setup, stack	setup guide; CI per commit	README ci.yml
④RK-05 demo data	deterministic seed, regression-tested	seed test
⑤RK-09, RK-10 hosting; personal data	accepted 9 Oct (risk plan); NFR-03 governs in-app access	risk plan

4. System models

Order items reference one product; orders belong to one user.

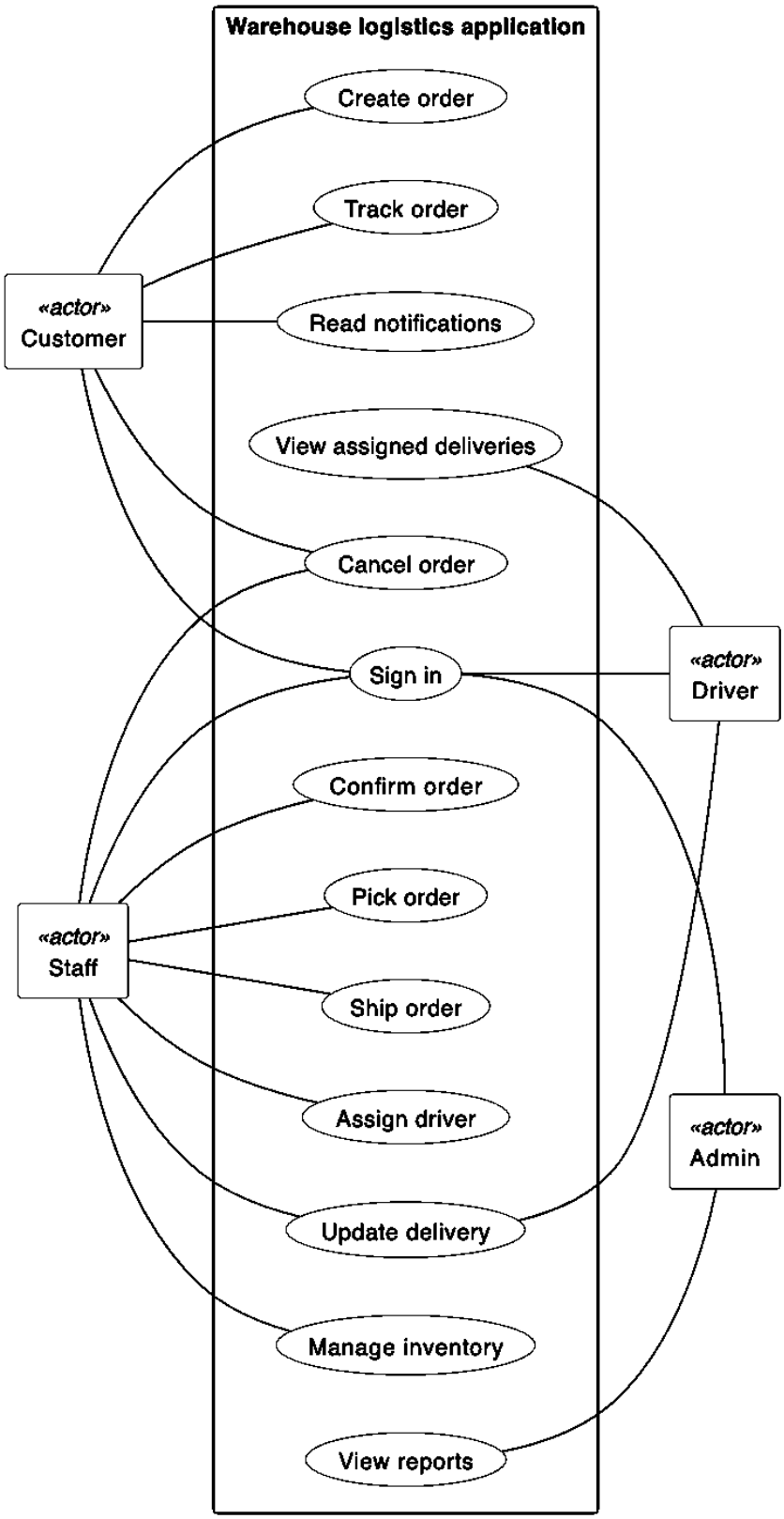


Figure 1.

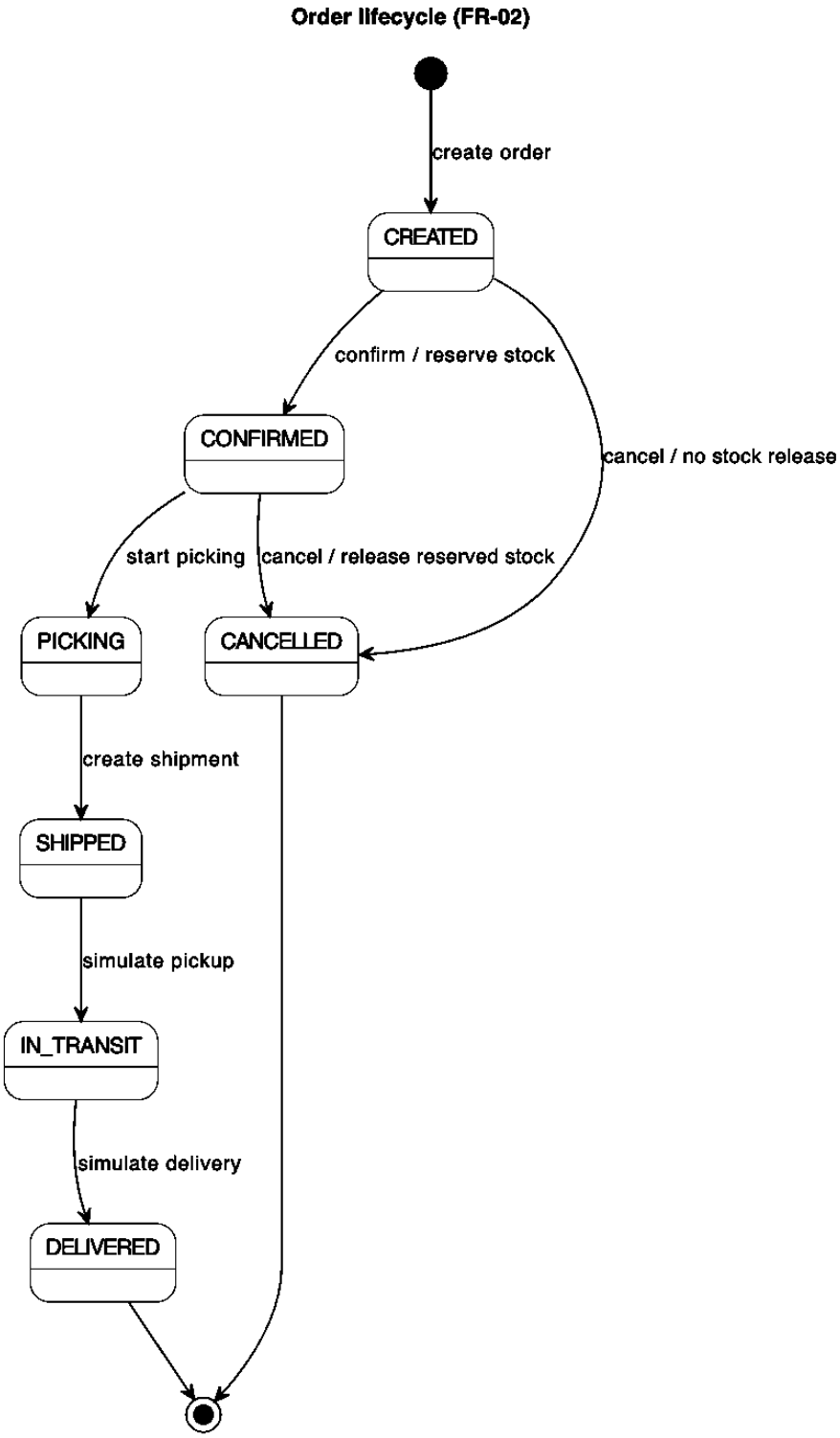


Figure 2. Order transitions: which reserve stock, which release it, which leave it unchanged.

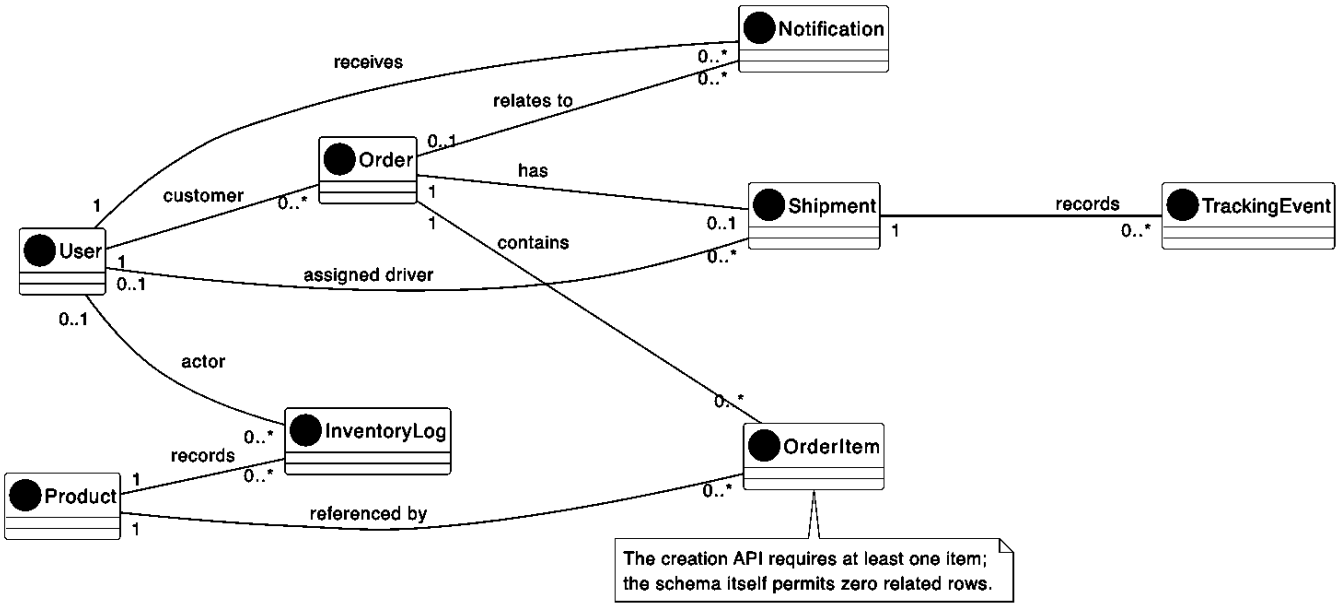


Figure 3.

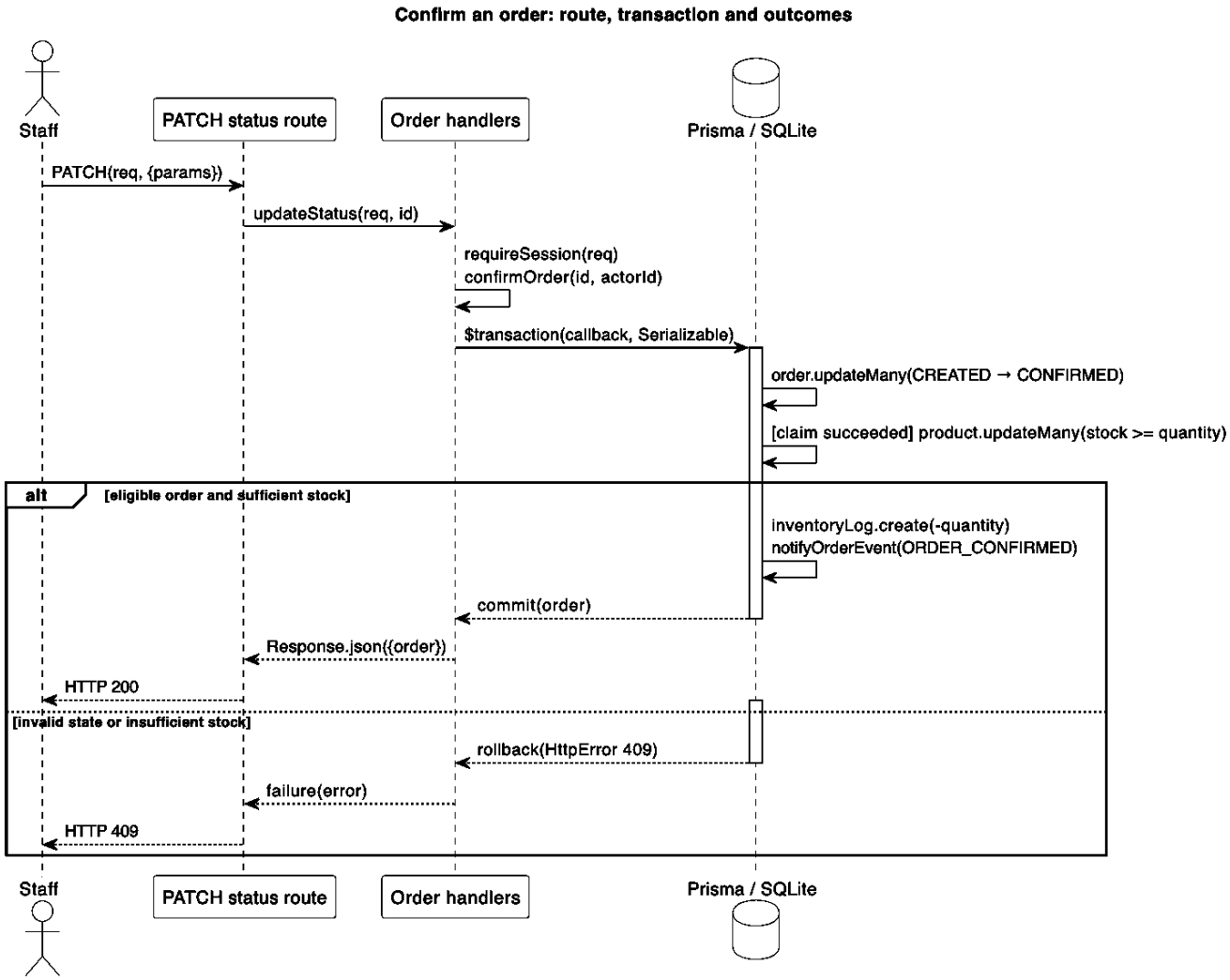


Figure 4. Confirmation: session guard, conditional writes, rollback.

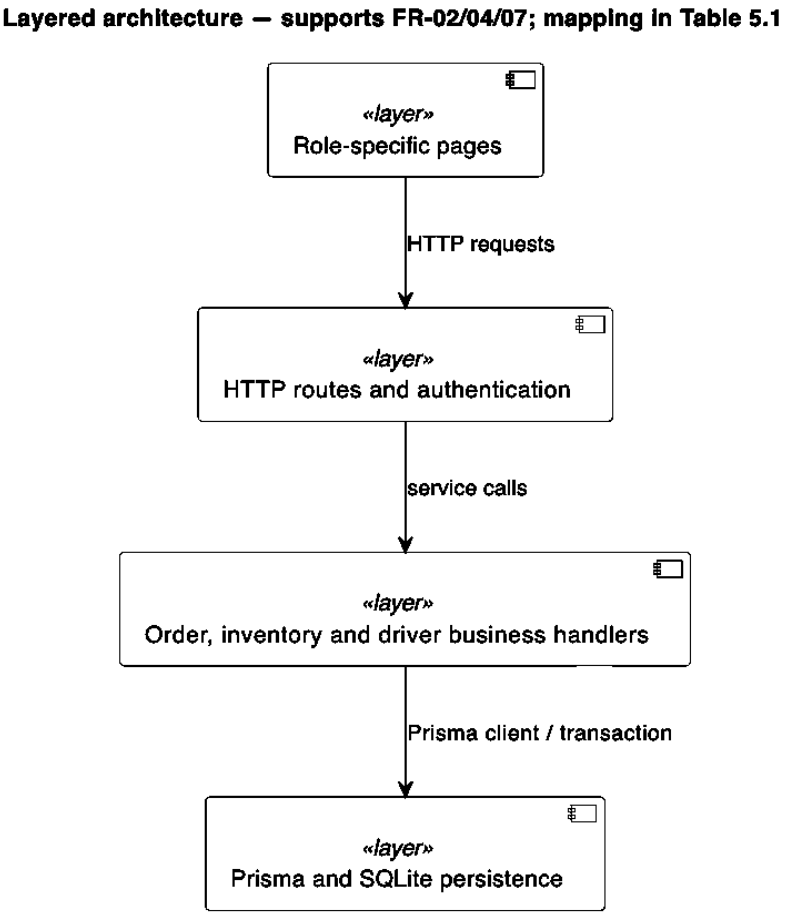


Figure 5.

5. Implementation

Confirmation reserves stock; lifecycle CREATED→CONFIRMED→PICKING→SHIPPED→IN_TRANSIT→DELIVERED; cancel from CREATED/CONFIRMED.

Cancellation permissions:

Role	Transition and effect
Customer	CREATED: cancel, no restock
Customer	CONFIRMED: cancel, restock
Staff	CREATED: rejected
Staff	CONFIRMED: cancel, restock
Customer, staff	PICKING onwards: rejected

A CREATED order awaits the customer's cancel or the staff confirmation; staff have no refuse action.

Atomic confirmation (HTTP 409 on repeats; rollback on short stock); driverId set at SHIPPED, immutable after; FR-03/04 verified (order-api, driver-api, inventory-api).

6. Verification

Verification checks the implementation against §3; validation exercises the delivered workflows (§8). CI records 38 unit, 140 integration, 2 integrity checks: the demo-data reset requires RESET_DEMO_DATA=YES (evidence protection), and the packaged submission's SHA-256 manifest must match the source (integrity checks).

Loopback measurement: p95 3.2–13.6 ms, first contentful paint p95 64–96 ms (30 samples per endpoint) vs 2/3 s — loopback, not representative of public networks.

Three reviews (rows 2, 6, 7): rows 2 and 6 verified requirements and models; row 7's fix was validated through the interface (§8).

7. Evolution

Reviews: 17 Oct found two defects (repeat submits, missing ledger) fixed by the confirmation transaction (§5); 10–11 Nov aligned diagrams and NFR-01/02 to CI-run checks; 11 Nov added direct export (issue #15). Feedback that the CI record was untested added two regression tests (row 8; suite 36→38).

8. Validation

Walkthrough covers eleven core operations: create order; open details; confirm; start picking; ship; assign driver; record pickup; record delivery; filter report; export current; export selected dates — at most three interactions each; the 26 checks reach all thirteen of Figure 1's use cases (coverage map). "Browser acceptance passed: 26 checks" (browser log): 20 checks (five operations once, three twice, three thrice) plus 6 cross-cutting (two each). During report preparation, 2 validators each attempted 32 core tasks: 64 of 64 attempts succeeded unaided, scored by the two recorders Chung Ngon Man and Li Mu Qiu; median 2.1 min (user-validation).

9. Feedback and reconciliation

Sources: whole-team reviews; test runs and the scripted walkthrough executed by Chung Ngon Man and Li Mu Qiu (Testing).

Table C. Feedback

Feedback (source, date)	Disposition and evidence
Stock effects moved from FR-04 to FR-02 (17 Oct, traceability)	—
Thin triggers (10–11 Nov, register)	Rewritten
Filter-export: four interactions (11 Nov, issue #15)	Combined path rejected — its steps already totalled four; direct path meets NFR-04; issue closed
No CI-health test (11 Nov, unit.log)	Two regression tests (row 8); suite 38
NFR-01/02 assumed hosted measurement, daily window (10–11 Nov, revision log)	CI-run checks

10. Evidence

38 unit, 140 integration, 2 integrity (12 Nov), 26 browser checks (11 Nov) — the only changes since were tests (history) — BASELINE. Traceability.